

HUMAN ENERGY ECONOMICS (HEE)

A Management Theory and Framework for Understanding, Managing, Recovering, and Investing in Human Energy



“

Organizations create Sustainable Business Value not by consuming more Human Energy, but by **protecting**, **recovering**, and **investing** it.

”



Md Mozammel Hoque
Management Researcher



Human Energy Economics (HEE)
Version 1.0 • July 2026



PROTECT HUMAN ENERGY



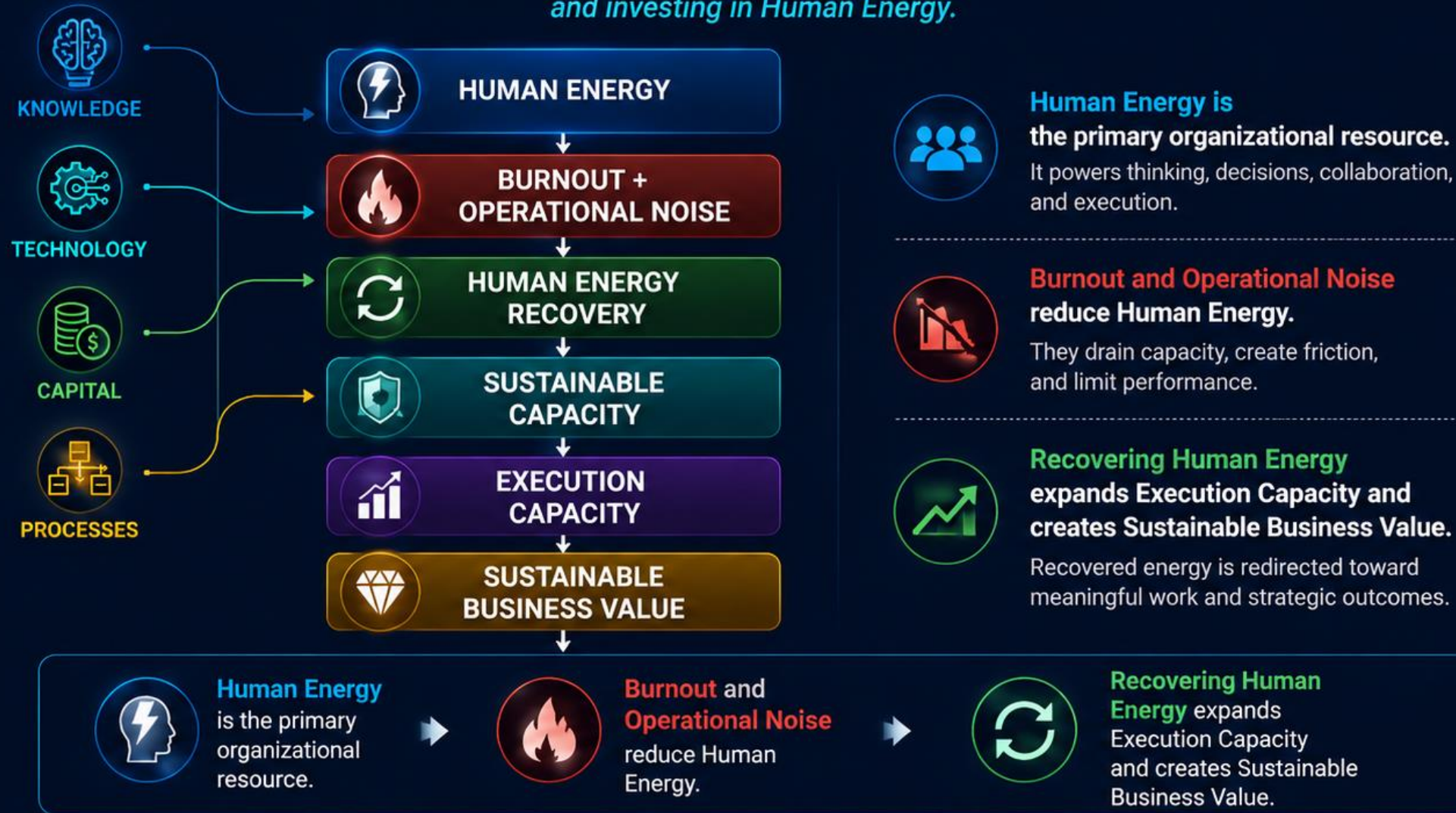
EXPAND EXECUTION CAPACITY



CREATE SUSTAINABLE BUSINESS VALUE

What is Human Energy Economics (HEE)?

A management theory for understanding, managing, recovering, and investing in Human Energy.



What Is Human Energy?

The Primary Organizational Resource That Creates Sustainable Business Value

HUMAN ENERGY CONSISTS OF



COGNITIVE CAPACITY

- Thinking
- Learning
- Problem solving
- Decision making



EMOTIONAL CAPACITY

- Motivation
- Resilience
- Collaboration
- Engagement



PHYSICAL CAPACITY

- Stamina
- Health
- Sustainable effort
- Recovery



KNOWLEDGE

Ideas • Expertise
Learning • Insight



TECHNOLOGY

Tools • Systems
Automation • Data



CAPITAL

Financial • Physical
Intellectual • Social



PROCESSES

Methods • Workflows
Policies • Practices



HUMAN ENERGY

(Cognitive • Emotional • Physical)



EXECUTION CAPACITY

Turning Potential into Performance



SUSTAINABLE BUSINESS VALUE

Long-term Value for All Stakeholders

HUMAN ENERGY IS



MEASURABLE

It can be assessed and tracked.



MANAGEABLE

It can be influenced and optimized.



RECOVERABLE

It can be restored and replenished.



INVESTABLE

It yields compounding returns.



THE PRIMARY RESOURCE

that converts every organizational asset into value.

“ Technology, knowledge, capital, and processes create value only when converted through **HUMAN ENERGY**. ”



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PROTECT HUMAN ENERGY



RECOVER CAPACITY



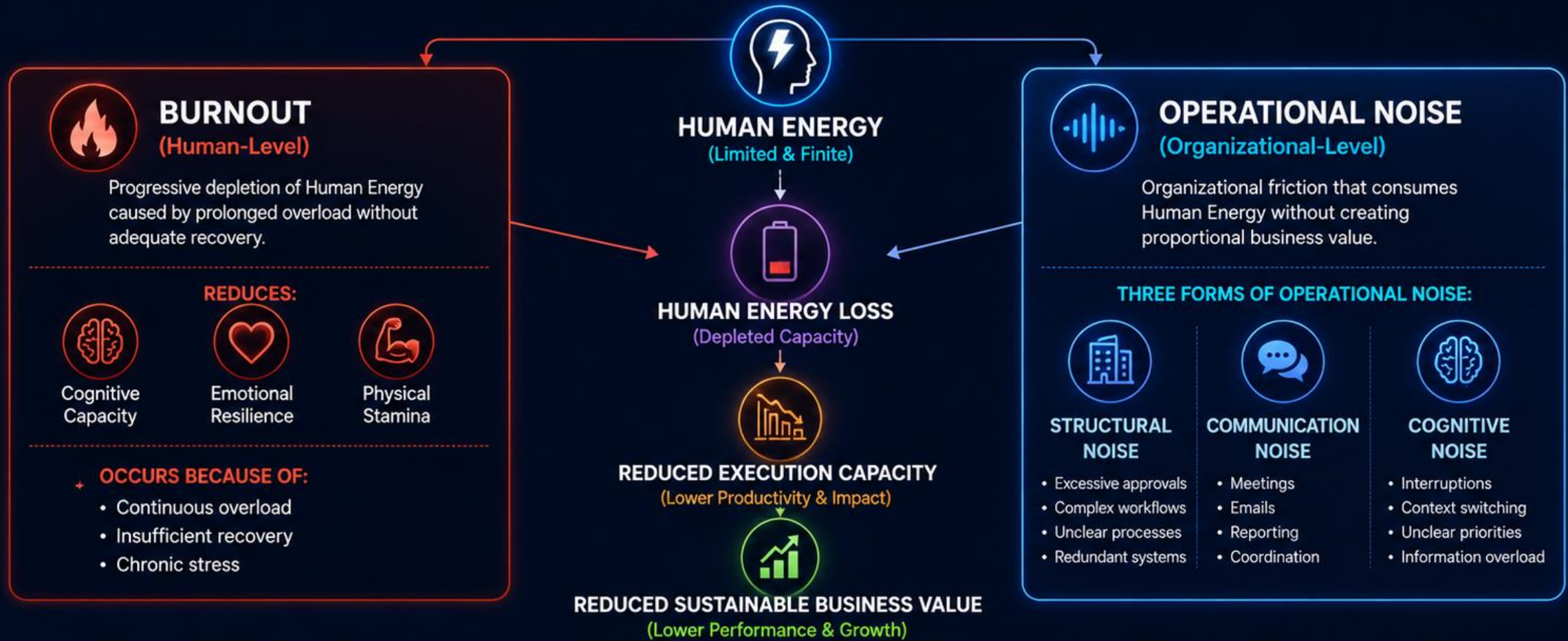
CREATE SUSTAINABLE BUSINESS VALUE



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BURNOUT & OPERATIONAL NOISE

The Two Primary Degraders of Human Energy



“ Human Energy is continuously degraded by **BURNOUT** at the individual level and **OPERATIONAL NOISE** at the organizational level. Together they reduce the Human Energy available for productive execution. ”



HUMAN ENERGY RECOVERY SYSTEM

How Organizations Recover Human Energy



OPERATIONAL SILENCE FRAMEWORK (OSF)



PURPOSE

Engineering Operational Silence by reducing recurring operational demand.



FOCUS

- ✓ Reduce Operational Noise
- ✓ Improve workflows
- ✓ Remove recurring interruptions
- ✓ Create sustainable operations
- ✓ Build people infrastructure



The goal is not to make people respond faster, but to eliminate the reasons they need to respond.



BURNOUT + OPERATIONAL NOISE



HUMAN ENERGY LOSS

(Depleted Capacity)



HUMAN ENERGY RECOVERY SYSTEM



RECOVERED HUMAN ENERGY

(Restored Capacity)



SUSTAINABLE CAPACITY

(Stable & Reliable)



EXECUTION CAPACITY

(High Performance)



SUSTAINABLE BUSINESS VALUE

(Impact & Growth)

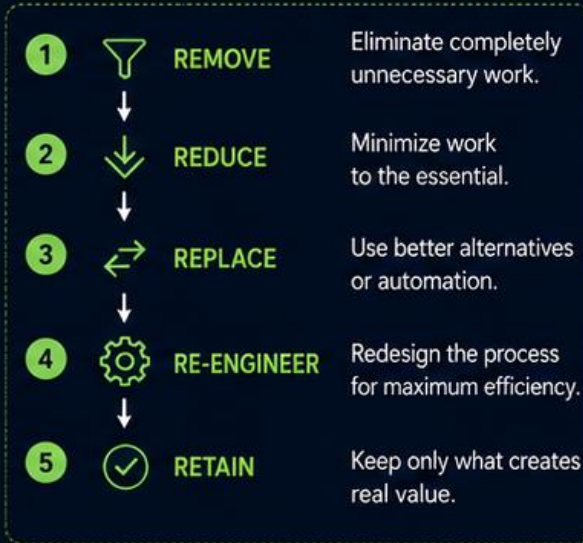


5R CASCADE FRAMEWORK



PURPOSE

A tactical decision methodology for eliminating unnecessary work and restoring Human Energy.



Human Energy Economics (HEE) explains **WHY** organizations should recover Human Energy.



Operational Silence Framework (OSF) explains **WHAT** organizational transformation looks like.



5R Cascade Framework explains **HOW** recurring operational demand is systematically eliminated.



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Management Researcher



RECOVER
HUMAN ENERGY



EXPAND
EXECUTION CAPACITY



CREATE
SUSTAINABLE BUSINESS VALUE



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HUMAN ENERGY LIFECYCLE

How Human Energy Changes Throughout Organizational Work



“

Human Energy is not static. It continuously moves through cycles of creation, deployment, degradation, recovery, rebuilding, and utilization. Organizations that intentionally manage this lifecycle sustain higher **Execution Capacity** and create greater **Sustainable Business Value**.

”



**PROTECT
HUMAN ENERGY**



**RECOVER
HUMAN ENERGY**



**EXPAND
EXECUTION CAPACITY**



**CREATE
SUSTAINABLE
BUSINESS VALUE**



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HUMAN ENERGY ECONOMICS CAPACITY MODEL

How Human Energy Is Converted into Sustainable Business Value



KEY EQUATIONS



$$HE = EC \times I \times DU$$



$$HEL = \text{Burnout} + \text{Operational Noise}$$



$$RHE = HE - HEL$$



$$SC = RHE \times AB \times CF \times AF \times XF$$



$$TSC = H \times SC$$



$$SBV = f(TSC, \text{Strategy}, \text{Market})$$

- EC = Energy per Capita
- I = Intensity
- DU = Duration of Utilization
- AB = Ability
- CF = Competence
- AF = Alignment
- XF = Experience Factor
- H = Number of People
- f() = Function

“

Human Energy is not valuable simply because it exists.
 It creates **Sustainable Business Value** only when protected, recovered,
 strengthened into **Sustainable Capacity**, and converted into effective execution.

”



PROTECT HUMAN ENERGY
 Reduce risk and prevent loss.



RECOVER HUMAN ENERGY
 Eliminate noise and recover lost energy.



BUILD SUSTAINABLE CAPACITY
 Strengthen ability, competence, alignment, and experience.



EXPAND EXECUTION CAPACITY
 Scale organization-wide execution capability.

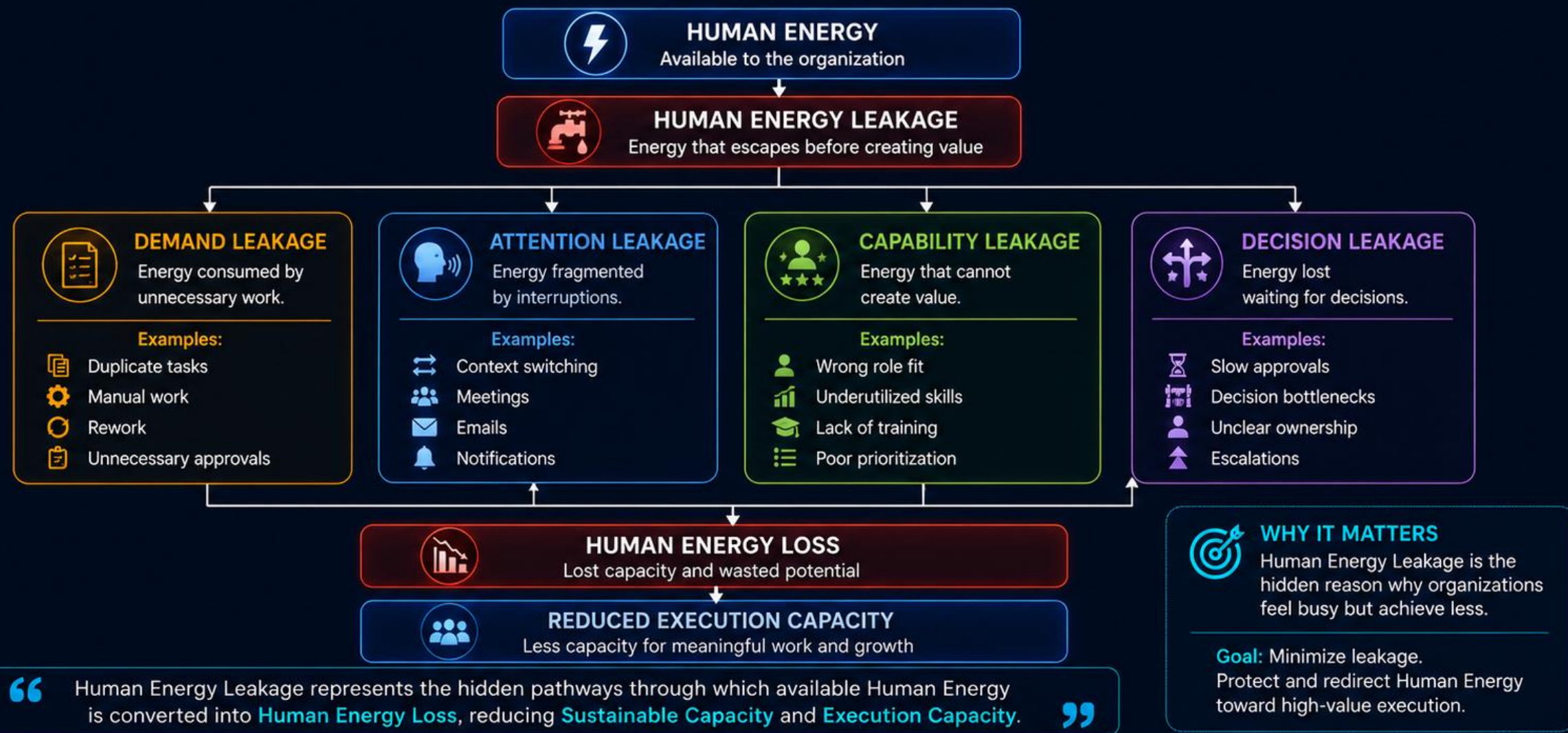


CREATE SUSTAINABLE BUSINESS VALUE
 Deliver long-term value through sustainable execution.

HUMAN ENERGY LEAKAGE FRAMEWORK

The Invisible Loss of Organizational Capacity

How Human Energy escapes before it creates value.



IDENTIFY LEAKAGE

See where energy is escaping.



MEASURE LEAKAGE

Quantify and understand the impact.



ELIMINATE LEAKAGE

Remove, reduce, or re-engineer the cause.



RECOVER ENERGY

Free up energy for meaningful work.



EXPAND CAPACITY

Increase execution capacity and results.

THE FOUR LAWS OF HUMAN ENERGY ECONOMICS

The Foundational Principles of Human Energy Economics

The fundamental laws that explain how Human Energy creates Sustainable Business Value.



PROTECT
HUMAN ENERGY



REDUCE
OPERATIONAL
NOISE



PREVENT
HUMAN ENERGY
LEAKAGE



RECOVER
HUMAN ENERGY



EXPAND
EXECUTION
CAPACITY



CREATE
SUSTAINABLE
BUSINESS VALUE

“ Together, these four laws provide the theoretical foundation of Human Energy Economics, explaining how organizations protect, recover, and invest Human Energy to create Sustainable Business Value. ”

HUMAN ENERGY AUDIT (HEA)

Identifying Where Human Energy Is Lost and Recovered

Measure before you improve. Diagnose before you transform.

WHAT HEA DELIVERS



Visibility into Human Energy consumption and loss.



Identification of root causes of energy leakage.



Quantification of recovery opportunities and impact.



Prioritized actions for reducing loss and increasing capacity.



Stronger Execution Capacity and Sustainable Business Value.



HUMAN ENERGY AUDIT (HEA)

A structured assessment to measure, analyze, and improve how Human Energy is used across the organization.



1. HUMAN ENERGY AUDIT

Measures overall Human Energy, loss, recovery opportunities, and organizational capacity.

Focus areas

- Human Energy Availability
- Energy Loss Analysis
- Recovery Potential
- Execution Capacity Index



2. OPERATIONAL NOISE AUDIT

Identifies unnecessary operational demand, process friction, communication overload, and recurring work.

Focus areas

- Process Noise
- Demand Noise
- Communication Noise
- Work Complexity



3. DECISION ENERGY AUDIT

Evaluates decision delays, approvals, ownership, governance, and decision-related energy consumption.

Focus areas

- Decision Delays
- Approval Friction
- Ownership Clarity
- Governance Efficiency



4. CAPABILITY & ALIGNMENT AUDIT

Assesses skills, role fit, capability gaps, strategic alignment, and development priorities.

Focus areas

- Skills & Competence
- Role Fit & Utilization
- Capability Gaps
- Strategic Alignment

HEA CONTINUOUS IMPROVEMENT CYCLE



SCAN

Understand current state and collect data.



DIAGNOSE

Identify root causes and quantify impact.



ELIMINATE

Remove, reduce, or re-engineer the causes.



INVEST

Reinvest recovered energy into high-value execution.



RECOVERED HUMAN ENERGY

Energy reclaimed from noise, friction, and inefficiencies.



EXPANDED EXECUTION CAPACITY

More focus, better performance, and higher productivity.



SUSTAINABLE BUSINESS VALUE

Greater impact, growth, and long-term value creation.

KEY HEA METRICS



OSI
Operational Silence Index



HERI
Human Energy Recovery Index



DER
Demand Elimination Rate



ECI
Execution Capacity Index



ONS
Operational Noise Score



HES
Human Energy Score



Human Energy Audit translates Human Energy Economics into measurable organizational action by identifying where Human Energy is consumed, degraded, recovered, and reinvested.



HUMAN ENERGY RECOVERY SYSTEM

From Diagnosis to Sustainable Organizational Improvement

An integrated system for recovering Human Energy and expanding Execution Capacity.

THE SYSTEM IN ACTION



1. DIAGNOSE

Identify where Human Energy is lost.



2. FOCUS

Eliminate what doesn't create value.



3. RECOVER

Reclaim lost Human Energy.



4. EXPAND

Strengthen capacity and performance.



5. CREATE VALUE

Deliver Sustainable Business Value.



HUMAN ENERGY ECONOMICS (HEE)

WHY

- Defines Human Energy as the foundational organizational resource.
- Explains Burnout, Operational Noise, and Human Energy Leakage.
- Establishes the principles for sustainable performance.



OPERATIONAL SILENCE FRAMEWORK (OSF)

WHAT

- Reduces recurring operational demand.
- Minimizes Operational Noise.
- Creates an environment that protects Human Energy.



5R CASCADE FRAMEWORK

HOW

1. REMOVE
Eliminate unnecessary work.

2. REDUCE
Reduce what is not necessary.

3. REPLACE
Replace with better alternatives.

4. RE-ENGINEER
Redesign for maximum efficiency.

5. RETAIN
Retain what truly creates value.



RECOVERED HUMAN ENERGY (RHE)
Energy reclaimed from noise, friction, and waste.



SUSTAINABLE CAPACITY (SC)
Consistent ability to deliver over time.



EXECUTION CAPACITY (ECap)
Greater focus, speed, and effectiveness.



SUSTAINABLE BUSINESS VALUE (SBV)
Long-term value creation for all stakeholders.

SYSTEM BENEFITS



Protects Human Energy
Reduces burnout and prevents energy loss.



Eliminates Operational Noise
Reduces recurring demand and distractions.



Recovers Lost Energy
Reclaims capacity for meaningful work.



Expands Execution Capacity
Improves focus, speed, and performance.



Creates Sustainable Business Value
Drives growth, innovation, and long-term impact.



Human Energy Recovery is achieved by combining **Human Energy Economics (WHY)**, the **Operational Silence Framework (WHAT)**, and the **5R Cascade Framework (HOW)** into one integrated management system.



ONE THEORY
Human Energy Economics



ONE OPERATING MODEL
Operational Silence Framework



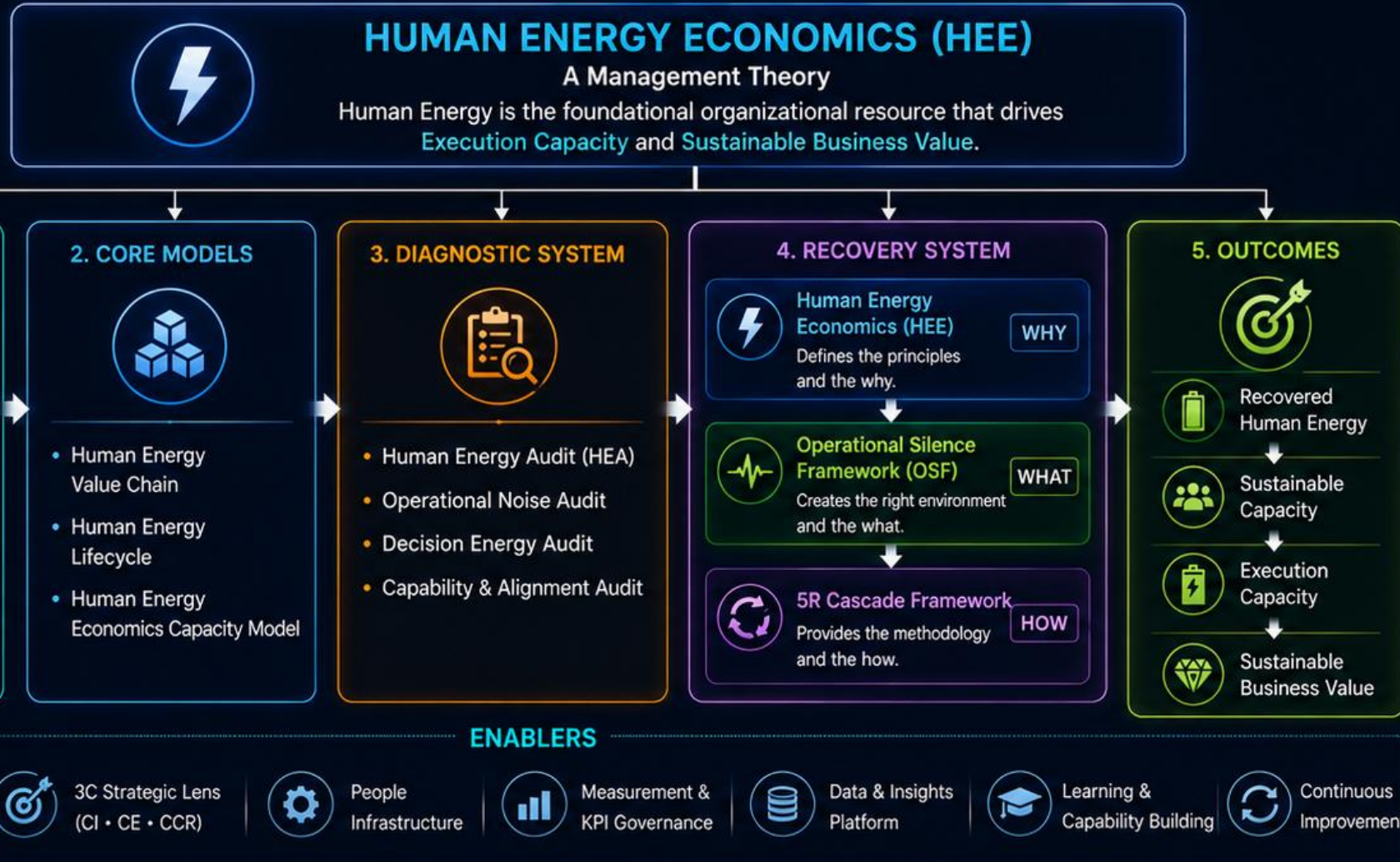
ONE DECISION FRAMEWORK
5R Cascade Framework



ONE HUMAN ENERGY RECOVERY SYSTEM
Protect • Recover • Expand • Create Value

HUMAN ENERGY ECONOMICS ARCHITECTURE

An Integrated Management Theory for Understanding, Managing, Recovering, and Investing in Human Energy



GOVERNING PRINCIPLES

- Protect Human Energy**
Eliminate the reasons people need to respond.
- Reduce Operational Noise**
Remove friction.
Reduce chaos.
- Prevent Human Energy Leakage**
Stop invisible losses of capacity.
- Recover Human Energy**
Reclaim lost capacity through smart actions.
- Expand Execution Capacity**
Convert recovered energy into useful execution.
- Create Sustainable Value**
Deliver long-term impact for all stakeholders.

“

Human Energy Economics provides an integrated management system that explains how Human Energy is **understood, measured, protected, recovered, and transformed** into **Sustainable Business Value**.

”

HUMAN ENERGY INVESTMENT CYCLE

Recovery Creates Capacity. Investment Creates Value.

THE PURPOSE OF HUMAN ENERGY RECOVERY

Recovering Human Energy is only the beginning.

Organizations create Sustainable Business Value by:



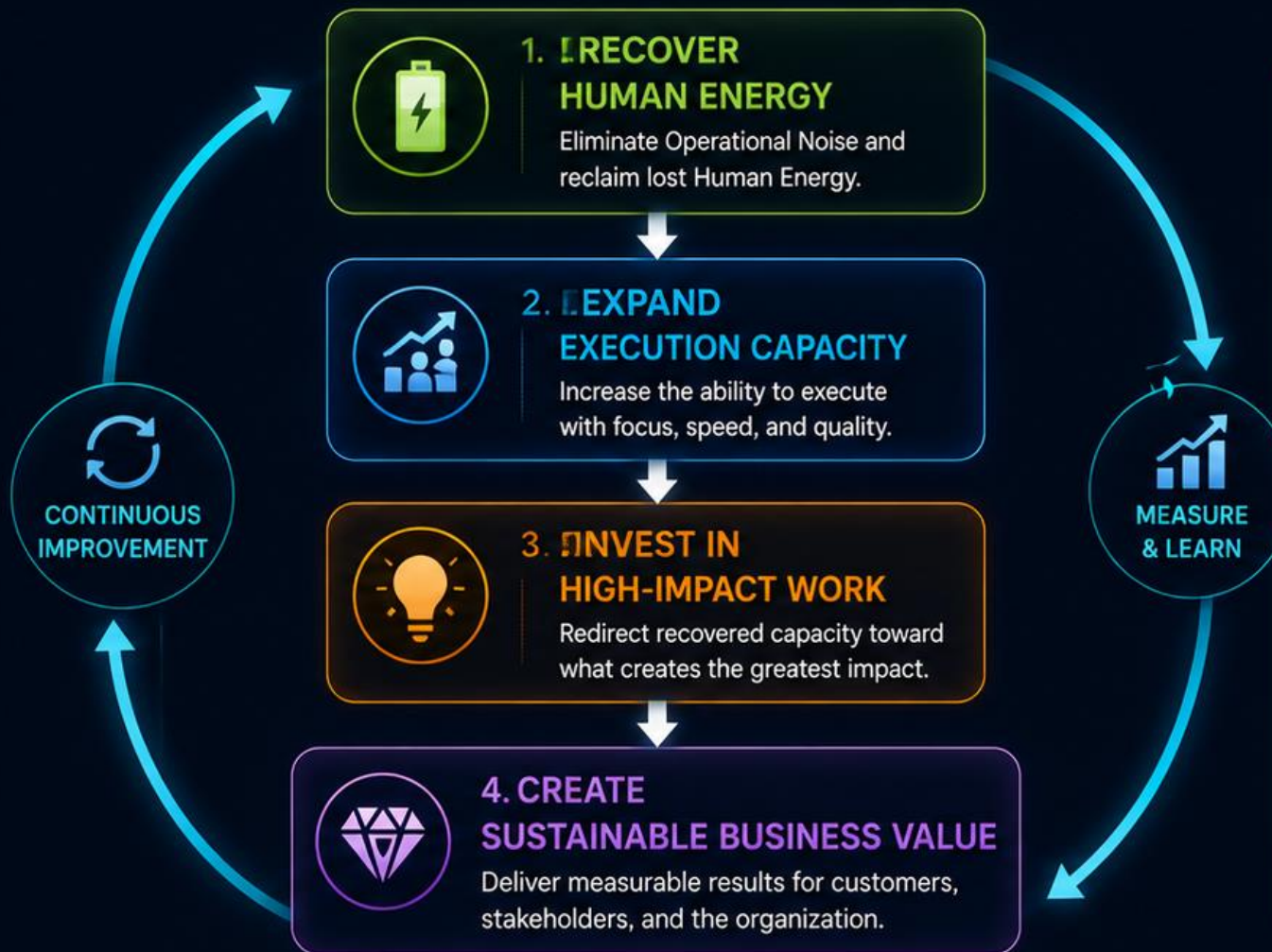
Recovering
Human Energy



Expanding
Execution Capacity



Redirecting capacity toward
high-impact work



INVESTMENT PRIORITIES



INNOVATION
Create new ideas, solutions, and competitive advantages.



STRATEGIC INITIATIVES
Advance priorities that drive long-term success.



CUSTOMER VALUE
Enhance experiences and deliver greater value.



CONTINUOUS IMPROVEMENT
Improve processes, systems, and ways of working.



ORGANIZATIONAL GROWTH
Develop people, capabilities, and culture for the future.

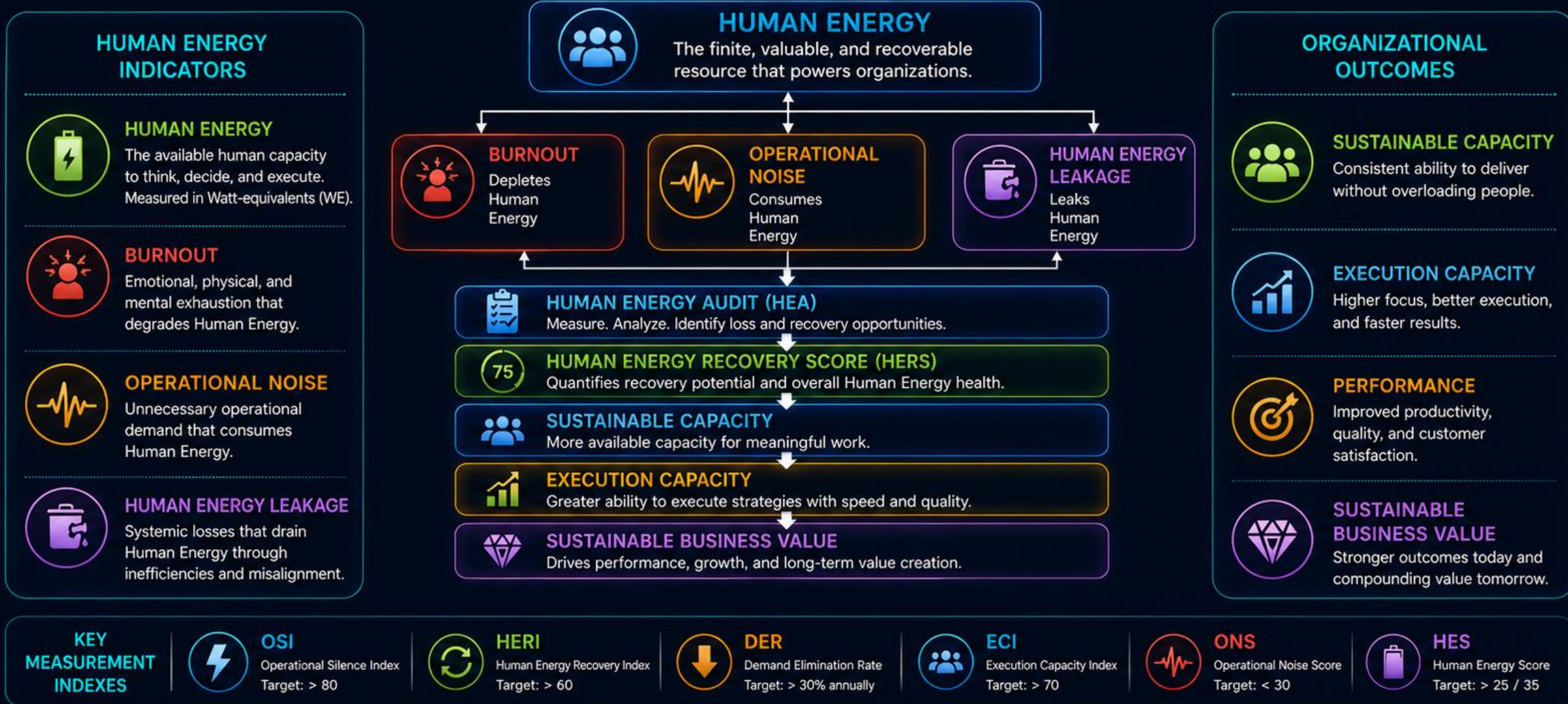
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Human Energy is **not** recovered to do more work.
It is recovered to do **more valuable work**.

”

MEASURING HUMAN ENERGY

What gets measured gets managed.



“

Organizations cannot improve what they do not measure.
Human Energy Economics begins by making Human Energy visible.

”

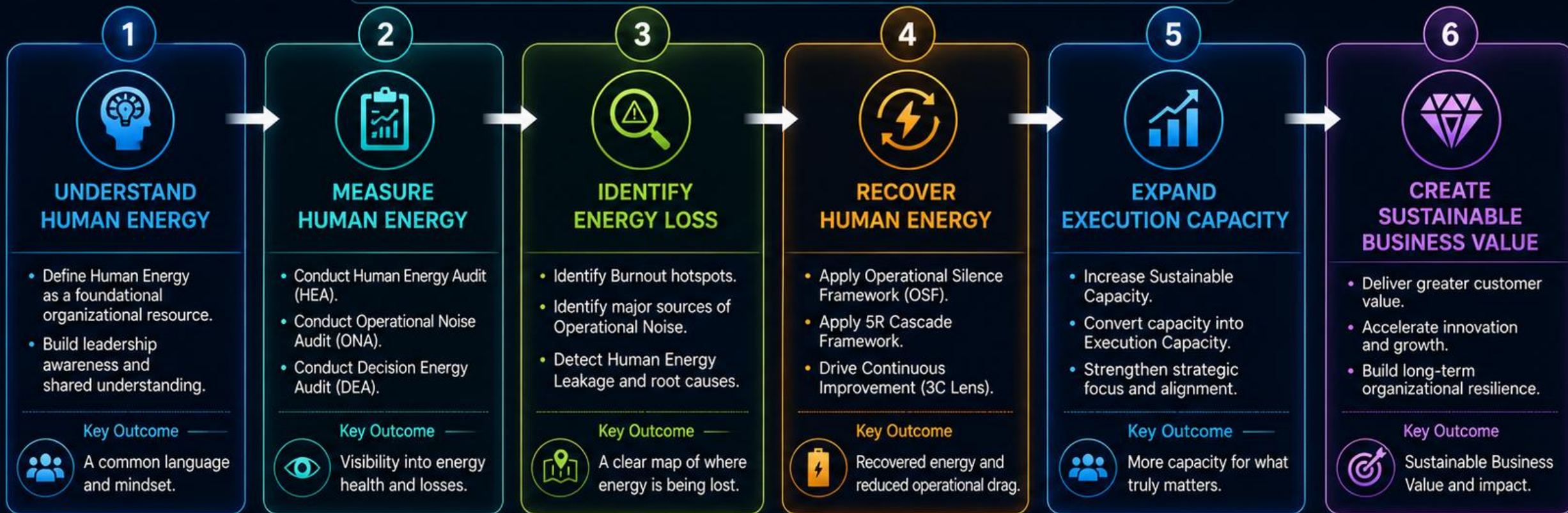
IMPLEMENTING HUMAN ENERGY ECONOMICS

A Practical Roadmap for Organizational Transformation

“

Human Energy Economics is not just a theory — it is a practical system that organizations can implement to achieve lasting results.

”



ENABLERS FOR SUCCESS



LEADERSHIP COMMITMENT
and Role Modeling



PEOPLE EMPOWERMENT
and Engagement



PROCESS SIMPLIFICATION
and Standardization



DATA & INSIGHTS
for Better Decisions
and Actions



GOVERNANCE & ACCOUNTABILITY
at Every Level



CAPABILITY BUILDING
and Learning



PROTECT



MEASURE



RECOVER



INVEST

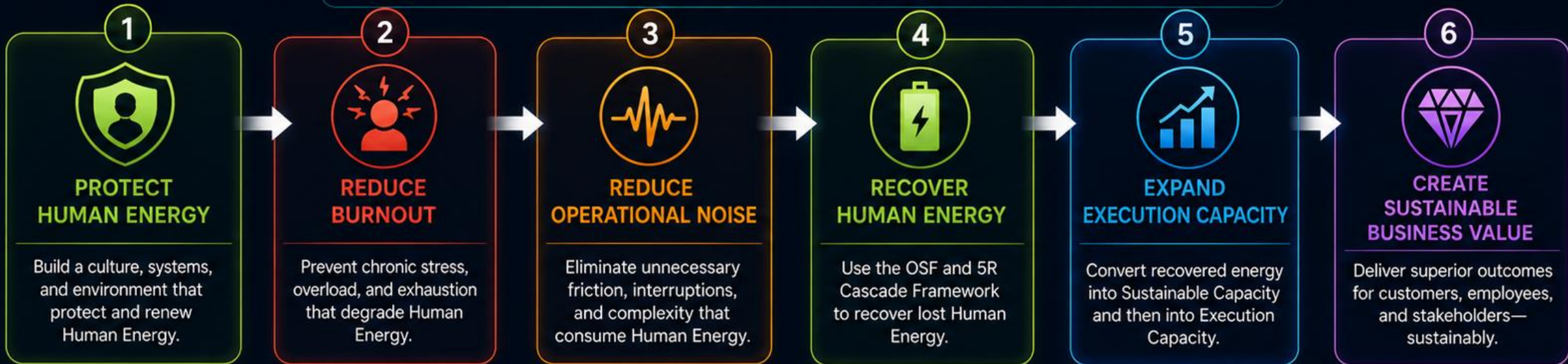


GROW

HUMAN ENERGY ECONOMICS IN ACTION

From Human Energy to Sustainable Business Value

“ Human Energy Economics transforms how organizations operate by **protecting**, **recovering**, and **investing** Human Energy. ”



THE OUTCOMES THAT MATTER



“ Organizations create sustainable value not by consuming more Human Energy, but by **protecting it**, **recovering it**, and **investing it** where it matters most. ”



WHY HUMAN ENERGY ECONOMICS MATTERS

The Future of Sustainable Organizational Performance

TRADITIONAL MANAGEMENT

Optimize people for work



MORE WORK

Increase demands and workload.



MORE PRESSURE

More stress, urgency, and firefighting.



MORE BURNOUT

People become exhausted and disengaged.



LOWER CAPACITY

Reduced focus, energy, and productivity.



DECLINING PERFORMANCE

Poor results, mistakes, rework, and lost opportunities.



Organizations do not
become high-performing
by **consuming more**
Human Energy.



They become high-performing
by **protecting**, **recovering**, and
investing Human Energy.



HUMAN ENERGY ECONOMICS

Optimize work for people



PROTECT HUMAN ENERGY

Build a healthy culture, systems, and environment.



REDUCE BURNOUT

Support well-being, balance, and sustainable pace.



REDUCE OPERATIONAL NOISE

Eliminate unnecessary work, friction, and interruptions.



RECOVER HUMAN ENERGY

Free up capacity through OSF and 5R Cascade Framework.



EXPAND EXECUTION CAPACITY

More focus, speed, quality, and strategic alignment.



CREATE SUSTAINABLE BUSINESS VALUE

Better results for customers, people, and the organization.

THE HUMAN ENERGY ECONOMICS PHILOSOPHY



PROTECT HUMAN ENERGY

It is our most valuable organizational resource.



REDUCE BURNOUT

Healthy people create sustainable performance.



ELIMINATE OPERATIONAL NOISE

Remove the work that should not exist.



RECOVER HUMAN ENERGY

Recover lost capacity and redirect it.



EXPAND EXECUTION CAPACITY

More capacity for what truly drives impact.



CREATE SUSTAINABLE BUSINESS VALUE

Enduring value for all stakeholders.

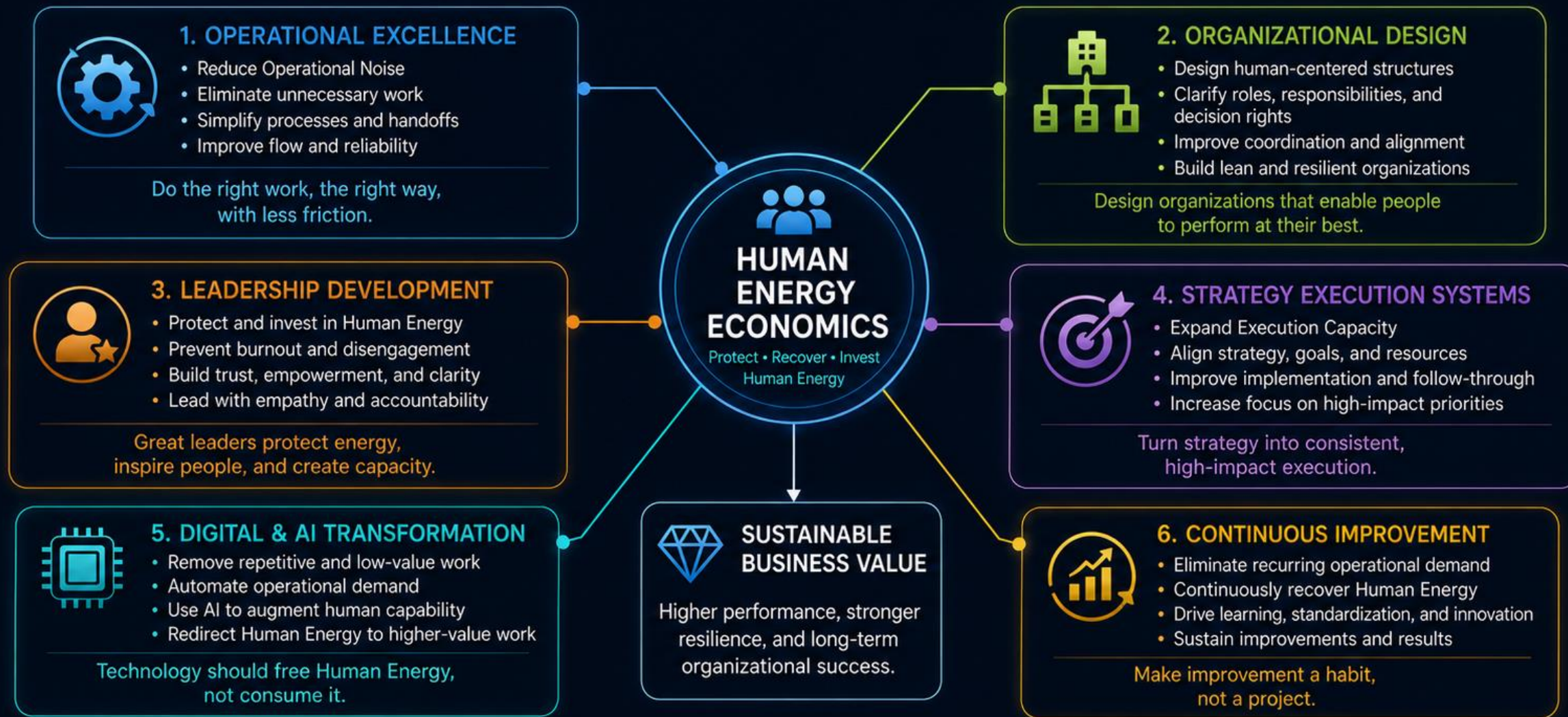
“

The organizations that win in the future will be those that **value**, **protect**, and **invest** in Human Energy — their people and their greatest potential.

”

APPLICATIONS OF HUMAN ENERGY ECONOMICS

Applying Human Energy to Improve Organizational Performance



“

Human Energy Economics provides a unifying management philosophy that enhances operational excellence, leadership, organizational design, digital transformation, and strategy execution by protecting, recovering, and investing in Human Energy.

”

KEY TAKEAWAYS

The Core Principles of Human Energy Economics



**PROTECT
HUMAN ENERGY**



**RECOVER
CAPACITY**



**EXPAND
EXECUTION**



**CREATE
SUSTAINABLE BUSINESS VALUE**

“

When we **protect** and **invest** in Human Energy, we **unlock** the greatest multiplier for performance, innovation, and value.

”

THE FUTURE OF MANAGEMENT

HUMAN ENERGY ECONOMICS

A New Management Paradigm

TRADITIONAL MANAGEMENT

- ✗ Optimize work
- ✗ Increase efficiency
- ✗ Improve productivity
- ✗ Scale people
- ✗ Respond faster

People are optimized for work.

THE HUMAN ENERGY ECONOMICS PHILOSOPHY



PROTECT HUMAN ENERGY



RECOVER CAPACITY



ENABLE BETTER EXECUTION



CREATE SUSTAINABLE BUSINESS VALUE

HUMAN ENERGY ECONOMICS

- ✓ Protect Human Energy
- ✓ Reduce Burnout
- ✓ Eliminate Operational Noise
- ✓ Recover Human Energy
- ✓ Expand Execution Capacity
- ✓ Create Sustainable Business Value



Work is optimized for people,
so people can create greater value.

“

Organizations do not achieve sustainable success by consuming more Human Energy. They achieve sustainable success by **protecting** it, **recovering** it, **investing** it wisely, and converting it into **Sustainable Business Value**.

”

“Human Energy is the invisible engine of every organization.

Protect it. Recover it. Invest it.
Everything else becomes possible.”



PROTECT
Human Energy



RECOVER
Capacity



ENABLE
Better Execution



CREATE
Sustainable Business Value

Human Energy Economics (HEE)

A Management Theory and Framework for Understanding, Managing, Recovering, and Investing in Human Energy



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THANK YOU

Protect Human Energy. Expand Execution Capacity. Create Sustainable Business Value.

HUMAN ENERGY IS THE INVISIBLE ENGINE OF EVERY ORGANIZATION.

Organizations do not achieve sustainable success by consuming more Human Energy.

They achieve sustainable success by:

- ✓ Protecting Human Energy
- ✓ Reducing Burnout
- ✓ Eliminating Operational Noise
- ✓ Preventing Human Energy Leakage
- ✓ Recovering Human Energy
- ✓ Expanding Execution Capacity
- ✓ Creating Sustainable Business Value



“

The future of management is not managing people harder.

*It is managing
Human Energy better.*

”



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